Statistical Foundations of Machine Learning

Solutions to Homework 1

September 8, 2026

Q 1

(a)

Each entry independently equals 1 with probability $\frac{1}{2}$. Same for 0.

1. $A_{2,2} = 1$:
$$\mathbb{P}(A_{2,2} = 1) = \frac{1}{2}$$

2. Given $A_{2,2} = 1$, the sum of the 2nd row equals 4 $\iff$ exactly 3 of the remaining 4 entries in the row are 1:
$$\mathbb{P}\left(\sum_{j \neq 2} A_{2,j} = 3\right) = \binom{4}{3}\left(\frac{1}{2}\right)^4 = \frac{4}{16}$$

3. Given $A_{2,2} = 1$, the sum of the 2nd column equals 3 $\iff$ exactly 2 of the remaining 4 entries in the column are 1:
$$\mathbb{P}\left(\sum_{i \neq 2} A_{i,2} = 2\right) = \binom{4}{2}\left(\frac{1}{2}\right)^4 = \frac{6}{16}$$

By independence of the entries, the required probability is:

$$\mathbb{P} = \left(\frac{1}{2}\right) \times \left[\binom{4}{3}\left(\frac{1}{2}\right)^4\right] \times \left[\binom{4}{2}\left(\frac{1}{2}\right)^4\right] = \frac{1 \times 4 \times 6}{2^9} = \frac{24}{512} = \frac{3}{64}$$

(b)

Let S be a subset chosen uniformly at random from the power set of $\{1, 2, \ldots, 10\}$.

- Total number of subsets $= 2^{10} = 1024$.
- The product of elements in S is odd if and only if S contains only odd integers from $\{1, 3, 5, 7, 9\}$ and is non-empty.

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The number of non-empty subsets of $\{1, 3, 5, 7, 9\}$ is:

$$2^5 - 1 = 31$$

Thus, the required probability is:

$$\mathbb{P} = \frac{2^5 - 1}{2^{10}} = \frac{31}{1024}$$

(Note: If the empty set is included with the empty product defined as odd, the probability is $\frac{2^5}{2^{10}} = \frac{1}{32}$.)

Q 2

(a)

The events $A \cap B$ and $A \cap B^c$ are disjoint, and their union is A:

$$(A \cap B) \cup (A \cap B^c) = A \quad \text{with} \quad (A \cap B) \cap (A \cap B^c) = \emptyset$$

By additivity of probability:

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$$

Since A and B are independent, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Substituting this in:

$$\begin{aligned} \mathbb{P}(A \cap B^c) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) \\ &= \mathbb{P}(A)\big(1 - \mathbb{P}(B)\big) \\ &= \mathbb{P}(A)\mathbb{P}(B^c) \end{aligned}$$

Thus, A and B^c are independent.

(b)

Given $\Omega = \{1, 2, 3, 4, 5, 6, 7, 8\}$ with uniform probability $\mathbb{P}(D) = \frac{|D|}{8}$, and:

$$A = \{1, 2, 5, 6, 7, 8\}, \quad B = \{1, 3, 5, 6, 7, 8\}, \quad C = \{1, 2, 3, 4\}$$

(i) Conditional Independence given C: First, compute the intersections with C:

$$A \cap C = \{1, 2\}, \quad B \cap C = \{1, 3\}, \quad A \cap B \cap C = \{1\}$$

Since $|C| = 4$, the conditional probabilities are:

$$\mathbb{P}(A \mid C) = \frac{|A \cap C|}{|C|} = \frac{2}{4} = \frac{1}{2}$$

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$$\mathbb{P}(B \mid C) = \frac{|B \cap C|}{|C|} = \frac{2}{4} = \frac{1}{2}$$
$$\mathbb{P}(A \cap B \mid C) = \frac{|A \cap B \cap C|}{|C|} = \frac{1}{4}$$

Checking the product:

$$\mathbb{P}(A \mid C) \times \mathbb{P}(B \mid C) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = \mathbb{P}(A \cap B \mid C)$$

Therefore, A and B **are conditionally independent** given C.

(ii) Independence of A and B: Compute the marginal and joint probabilities:

$$|A| = 6 \implies \mathbb{P}(A) = \frac{6}{8} = \frac{3}{4}$$
$$|B| = 6 \implies \mathbb{P}(B) = \frac{6}{8} = \frac{3}{4}$$

The intersection is $A \cap B = \{1, 5, 6, 7, 8\}$, so $|A \cap B| = 5$:

$$\mathbb{P}(A \cap B) = \frac{5}{8}$$

Checking the independence condition:

$$\mathbb{P}(A)\mathbb{P}(B) = \frac{3}{4} \times \frac{3}{4} = \frac{9}{16}$$

Since $\mathbb{P}(A \cap B) = \frac{5}{8} = \frac{10}{16} \neq \frac{9}{16}$:

$$\mathbb{P}(A \cap B) \neq \mathbb{P}(A)\mathbb{P}(B)$$

Therefore, A and B **are not independent**.

Q 3

(a)

Events of getting exactly one six: $6aa, a6a, aa6$ where a denotes a number that is not 6

$$\mathbb{P}(\mathcal{E}_1) = 3 \left(\frac{1}{6}\right)^1 \left(\frac{5}{6}\right)^2 = 3 \times \frac{25}{216} = \frac{75}{216} = \frac{25}{72}$$

(b)

Using the complement (the probability that no six appears in any of the three rolls):

$$\mathbb{P}(\mathcal{E}_2) = 1 - \mathbb{P}(\text{no } 6) = 1 - \left(\frac{5}{6}\right)^3 = 1 - \frac{125}{216} = \frac{91}{216}$$

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(c)

Let A_i denote the event that a 6 appears on the ith roll, for $i \in \{1, 2, 3\}$. Then $\mathcal{E}_2 = \bigcup_{i=1}^{3} A_i$.

By independence:

$$S_1 = \sum_{i=1}^{3} \mathbb{P}(A_i) = 3 \times \frac{1}{6} = \frac{1}{2} = \frac{108}{216}$$

$$S_2 = \sum_{1 \le i < j \le 3} \mathbb{P}(A_i \cap A_j) = \binom{3}{2} \left(\frac{1}{6}\right)^2 = 3 \times \frac{1}{36} = \frac{1}{12} = \frac{18}{216}$$

$$S_3 = \mathbb{P}(A_1 \cap A_2 \cap A_3) = \left(\frac{1}{6}\right)^3 = \frac{1}{216}$$

The Bonferroni inequalities give:

- **First-order bound (Upper bound):**

$$\mathbb{P}(\mathcal{E}_2) \le S_1 = \frac{1}{2} = \frac{108}{216} = 0.5$$

- **Second-order bound (Lower bound):**

$$\mathbb{P}(\mathcal{E}_2) \ge S_1 - S_2 = \frac{1}{2} - \frac{1}{12} = \frac{5}{12} = \frac{90}{216} \approx 0.4167$$

- **Third-order bound (Exact value):**

$$\mathbb{P}(\mathcal{E}_2) = S_1 - S_2 + S_3 = \frac{90}{216} + \frac{1}{216} = \frac{91}{216} \approx 0.4213$$

Comparison: The bounds satisfy:

$$S_1 - S_2 \le \mathbb{P}(\mathcal{E}_2) \le S_1 \iff \frac{90}{216} \le \frac{91}{216} \le \frac{108}{216}$$

The true value $\frac{91}{216}$ lies strictly between the second-order lower bound and the first-order upper bound, and coincides exactly with the third-order expansion since the union contains only 3 events.

Q 4

Step 1: Compute individual probabilities for single roll of a pair of dice The total number of outcomes is $6 \times 6 = 36$.

- **A rolls a total of 9:** Favorable pairs are $(3,6), (4,5), (5,4), (6,3)$ (4 outcomes).

$$p_A = \mathbb{P}(\text{A rolls 9}) = \frac{4}{36} = \frac{1}{9}, \quad q_A = 1 - p_A = \frac{8}{9}$$

- **B rolls a total of 4:** Favorable pairs are $(1,3), (2,2), (3,1)$ (3 outcomes).

$$p_B = \mathbb{P}(\text{B rolls 4}) = \frac{3}{36} = \frac{1}{12}, \quad q_B = 1 - p_B = \frac{11}{12}$$

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Step 2: Probability that B wins The order of throws is $A, B, A, B, \ldots$ with A starting. For B to win, B must roll a total of 4 before A rolls a total of 9.

B can win on their 1st turn, 2nd turn, 3rd turn, and so on:

$$\begin{aligned}\mathbb{P}(B \text{ wins}) &= q_A p_B + (q_A q_B) q_A p_B + (q_A q_B)^2 q_A p_B + \ldots \\ &= \frac{q_A p_B}{1 - q_A q_B}\end{aligned}$$

Substitute the values:

$$q_A p_B = \frac{8}{9} \times \frac{1}{12} = \frac{2}{27}$$

$$q_A q_B = \frac{8}{9} \times \frac{11}{12} = \frac{22}{27} \implies 1 - q_A q_B = 1 - \frac{22}{27} = \frac{5}{27}$$

Thus:

$$\mathbb{P}(B \text{ wins}) = \frac{\frac{2}{27}}{\frac{5}{27}} = \frac{2}{5}$$

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